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Lin

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(54) **SAFETY HOOK**

(75) Inventor: **Yu-Wen Lin**, Taipei (TW)

(73) Assignee: **Usang Industrial Co., Ltd.**, Taipei (TW)

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(30) **Foreign Application Priority Data**

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(51) **Int. Cl.**
F16B 45/02 (2006.01)

(52) **U.S. Cl.** **24/600.1**

(58) **Field of Classification Search** 24/599.5, 24/600.1; 294/82.2

See application file for complete search history.

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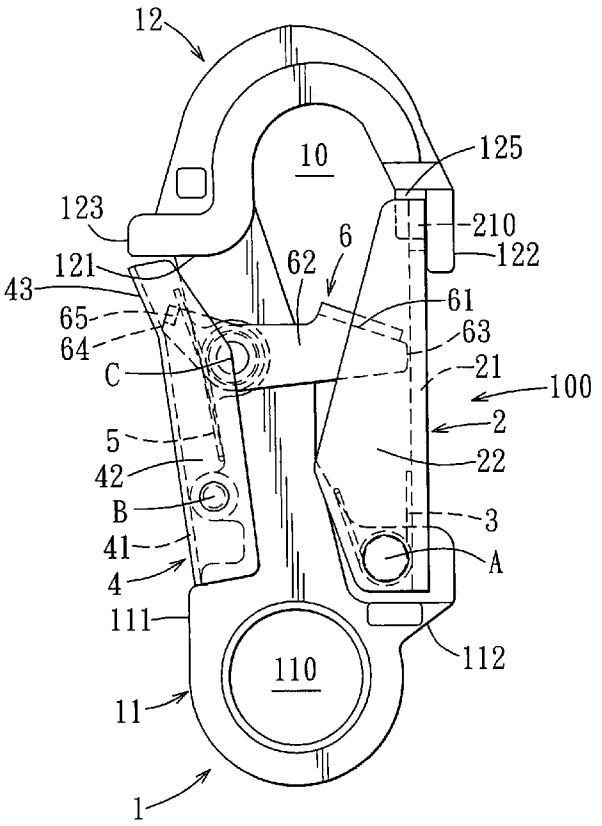
* cited by examiner

Primary Examiner—James R Brittain
(74) *Attorney, Agent, or Firm*—Harness, Dickey & Pierce, P.L.C.

(57) **ABSTRACT**

A safety hook includes: a hook body defining a hook mouth; a latch unit including a latch plate pivoted to the hook body so as to be rotatable between locked and unlocked positions, the latch plate having a blocking plate portion; a releasing unit including a releasing plate pivoted to the hook body, the releasing plate having a driving plate portion; and a safety lock unit including a bent safety plate disposed between the latch plate and the releasing plate, pivotable between restraining and releasing positions, and having a restraining end portion abutting against the blocking plate portion, and a driven end portion abutting against the driving plate portion.

5 Claims, 7 Drawing Sheets



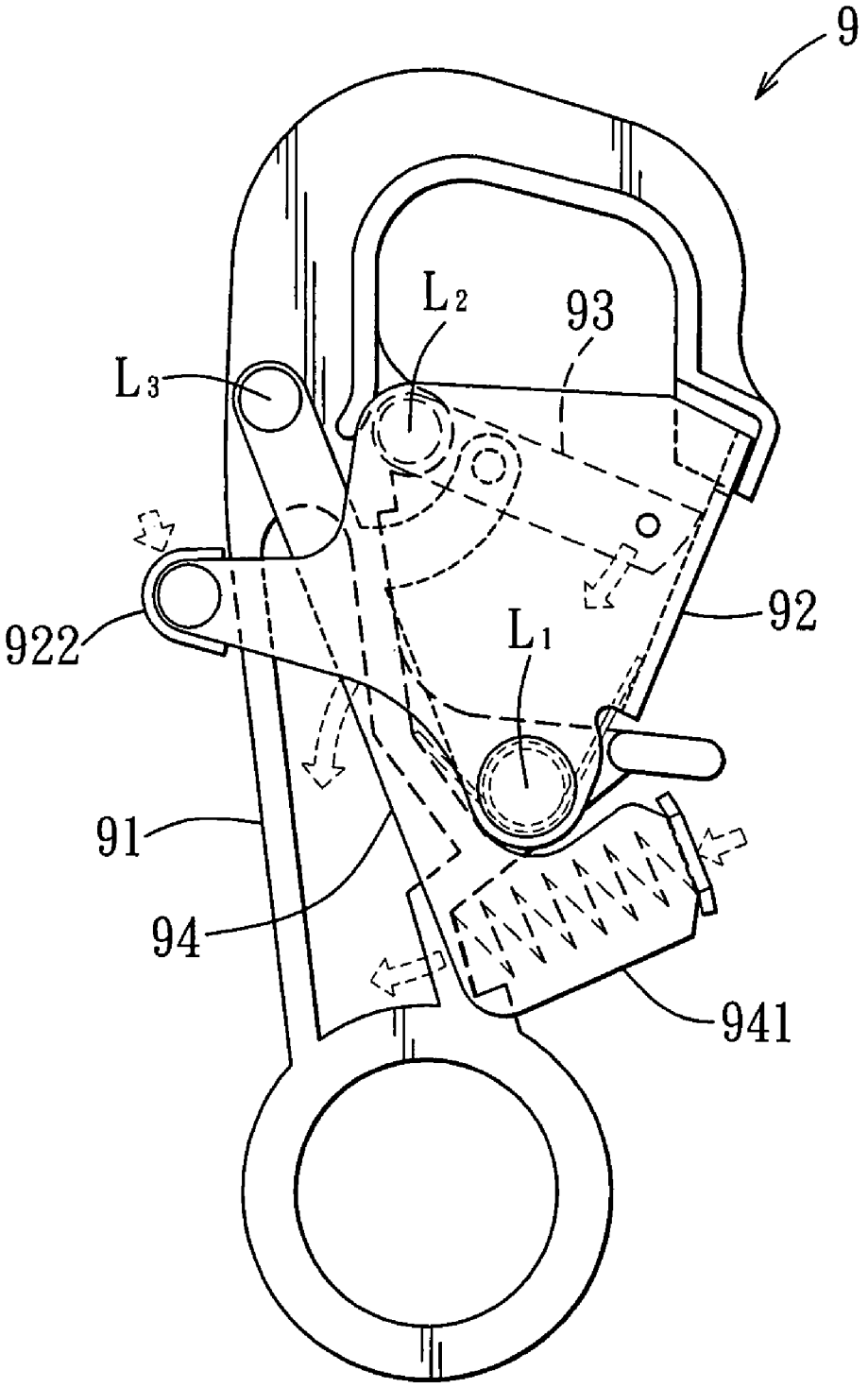


FIG. 1
PRIOR ART

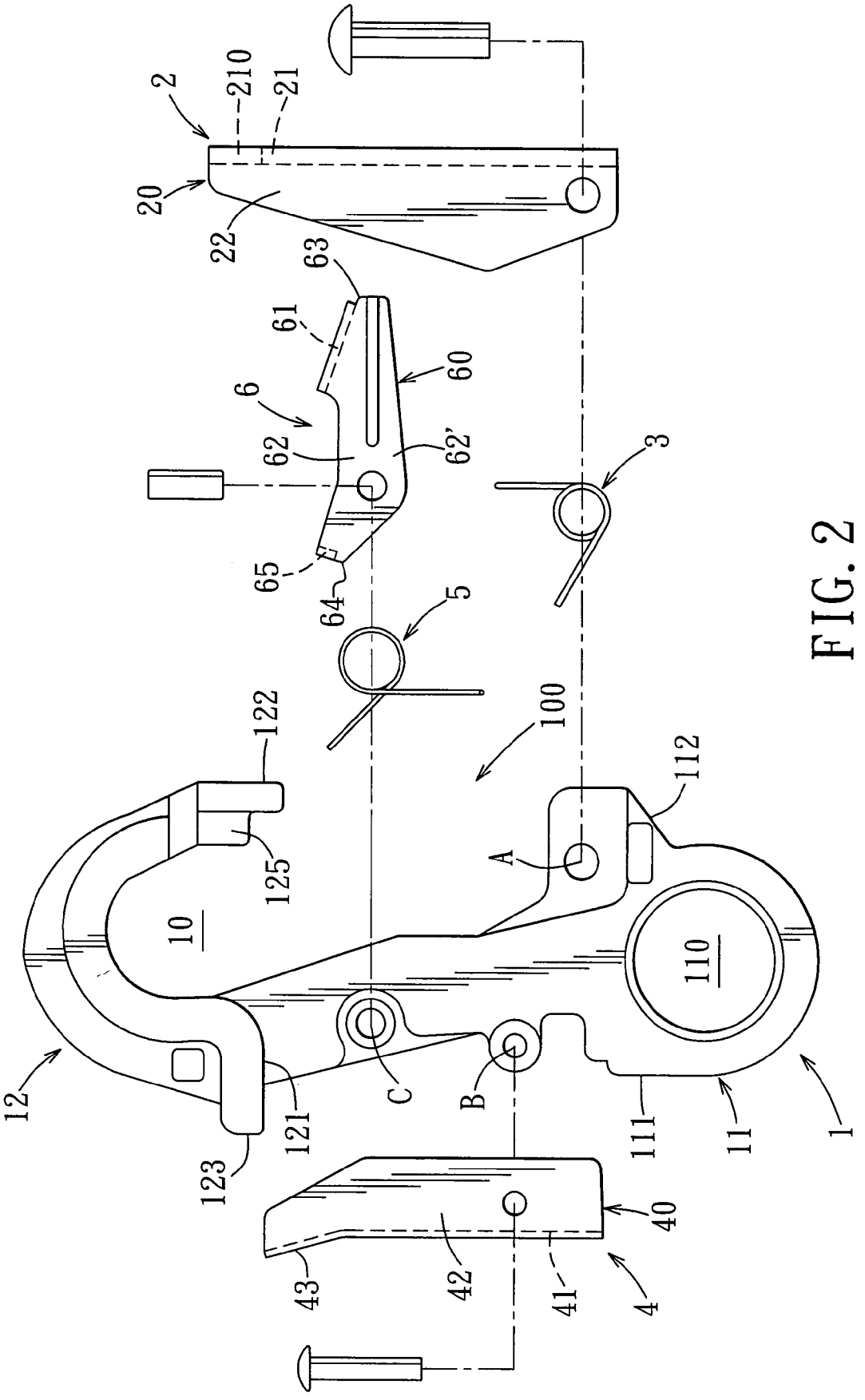

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FIG. 2

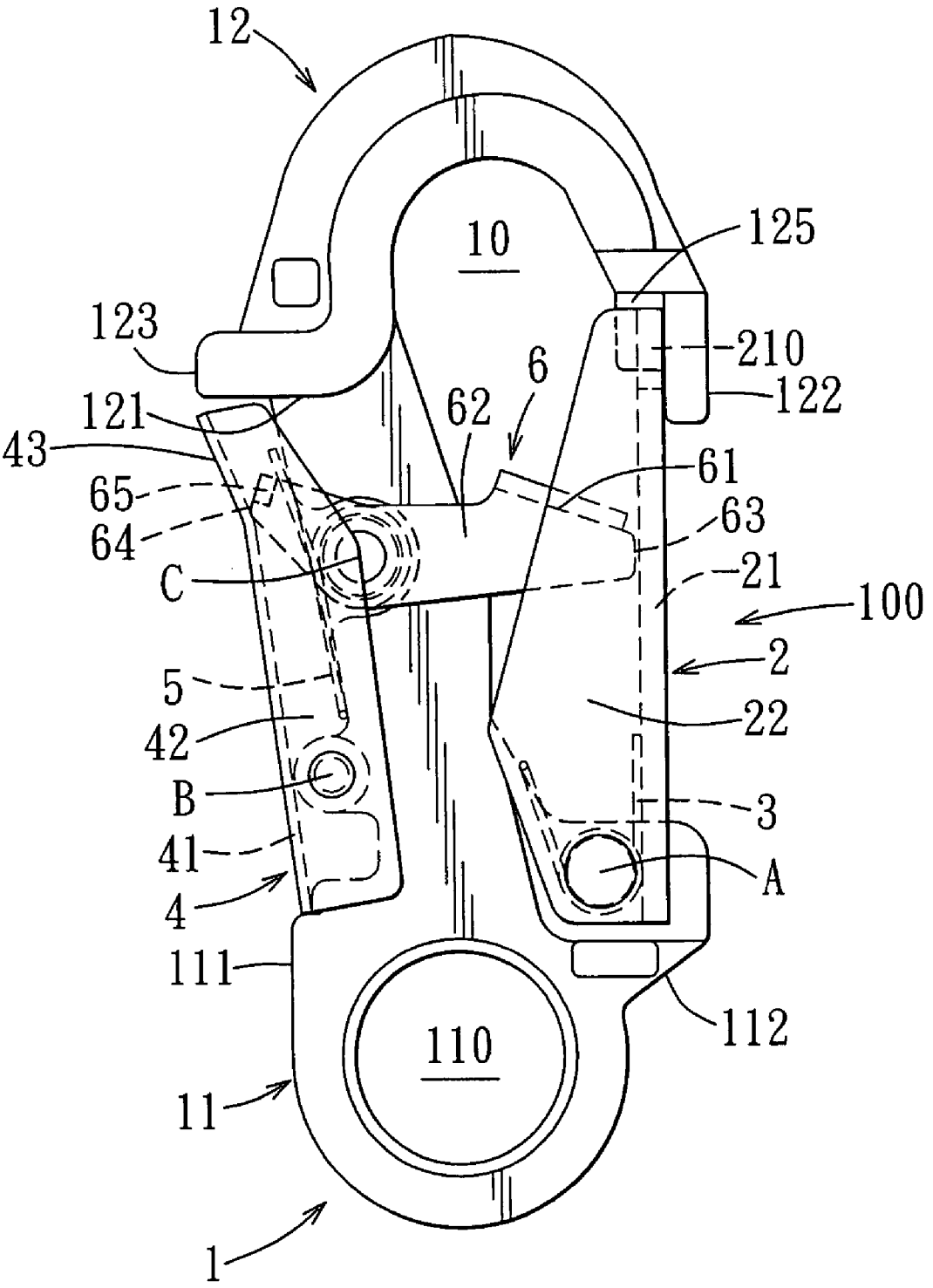

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FIG. 3

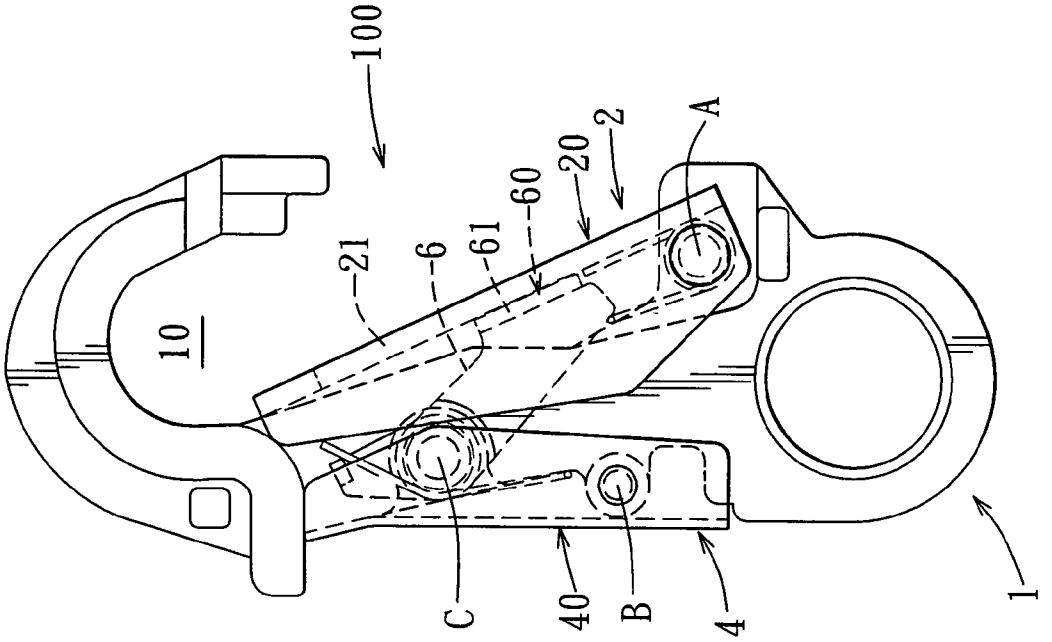


FIG. 5

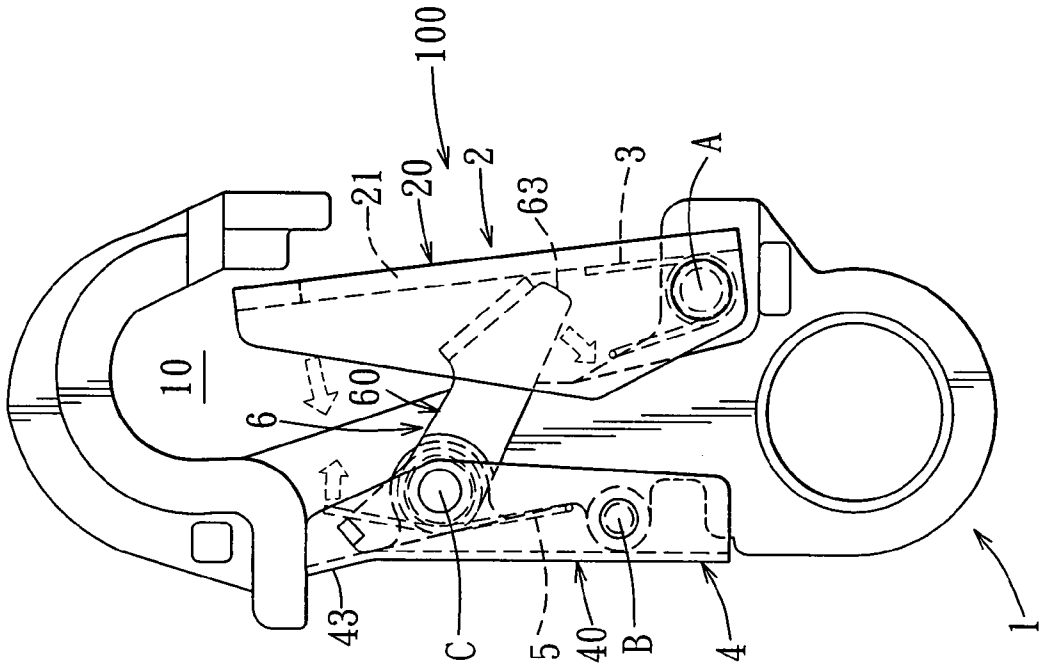


FIG. 4

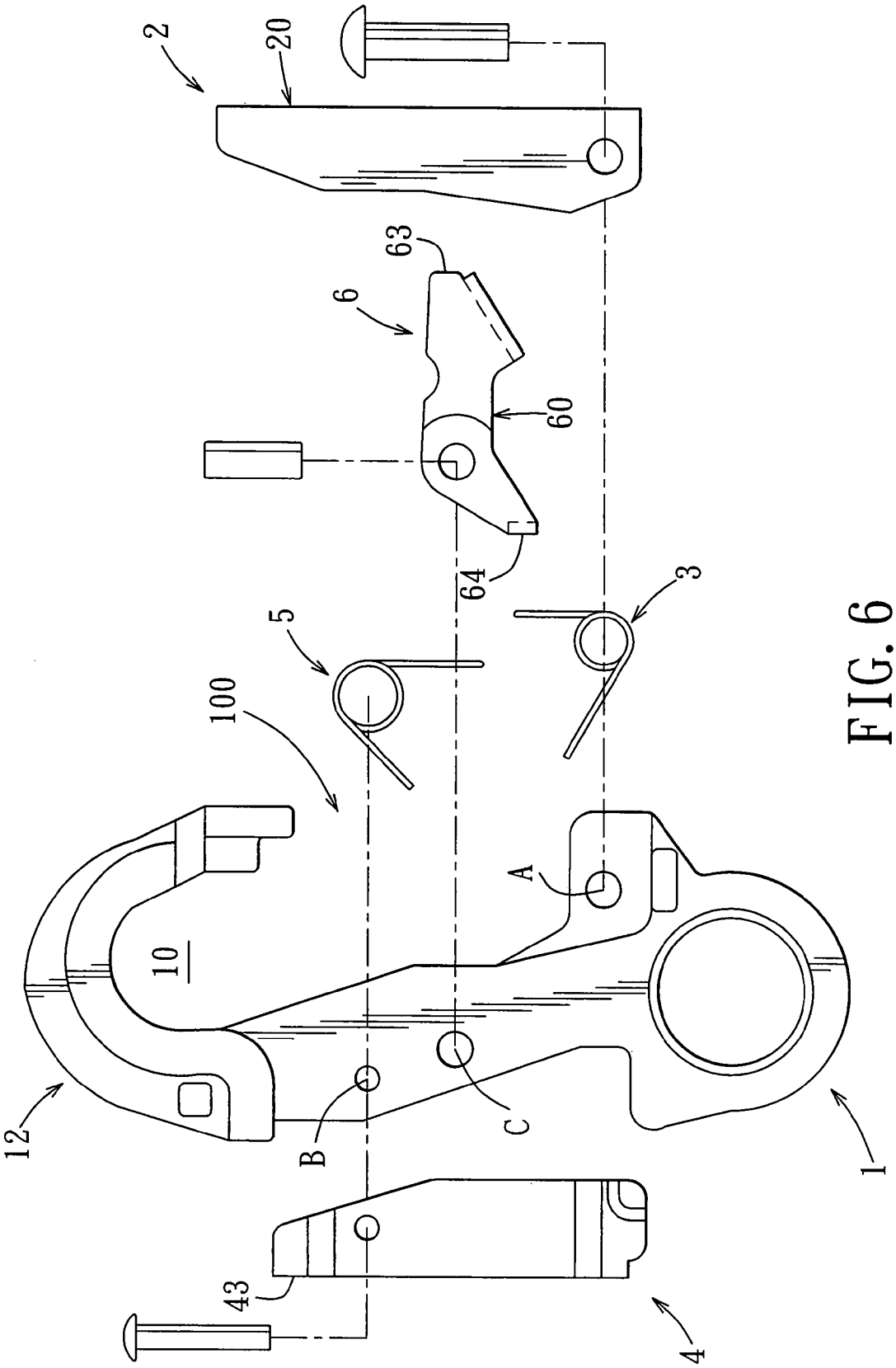


FIG. 6

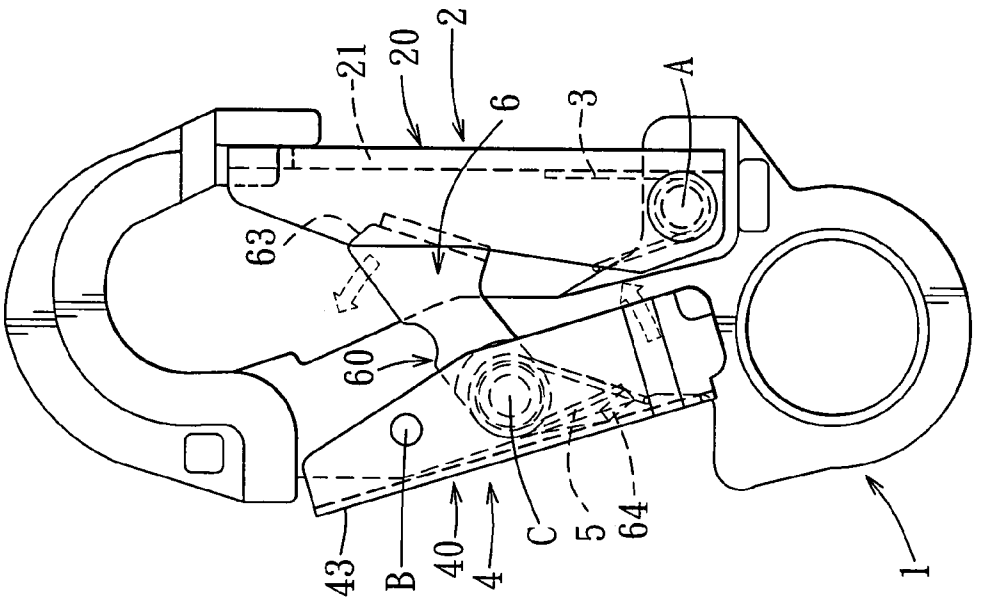


FIG. 8

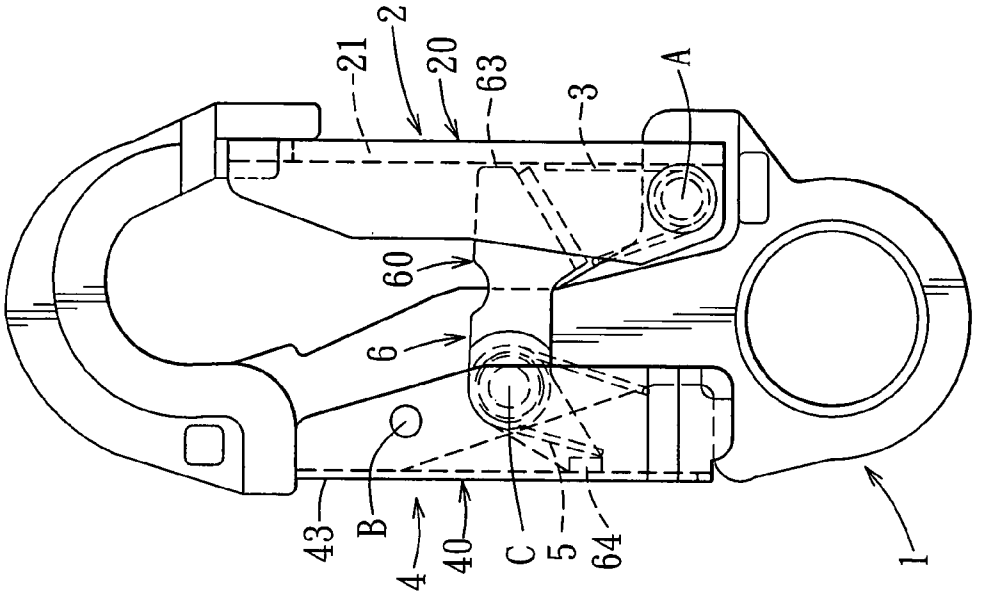


FIG. 7

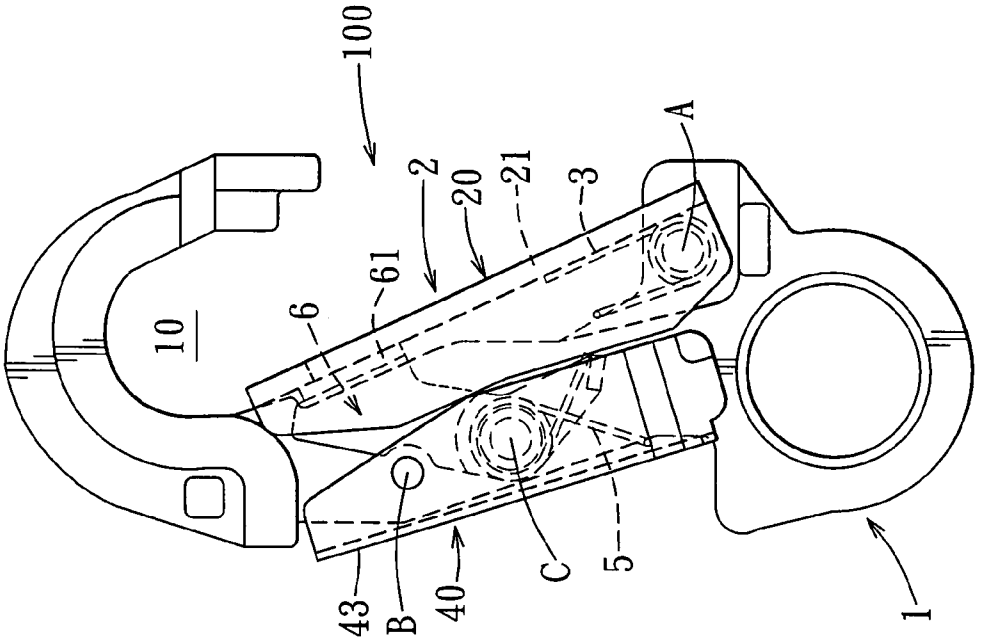


FIG. 10

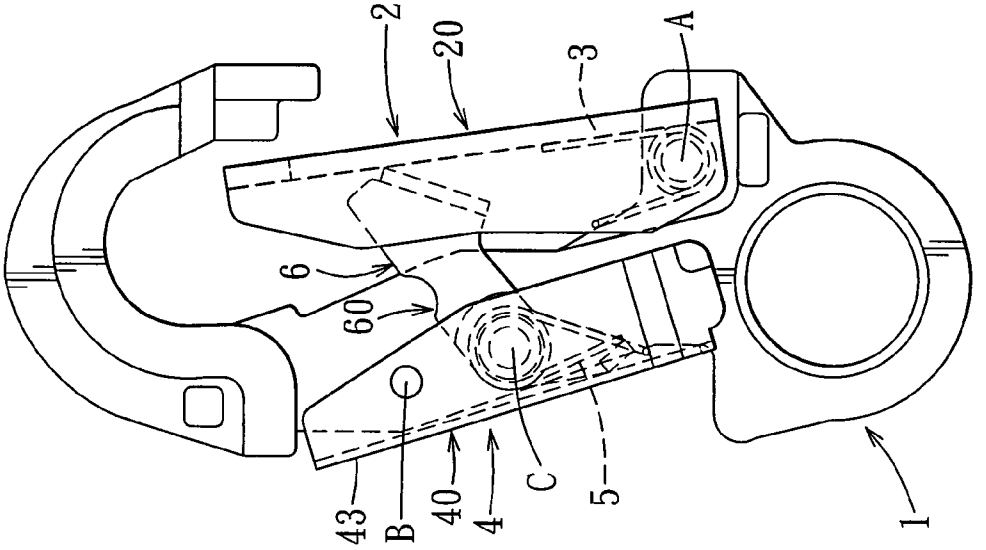


FIG. 9

SAFETY HOOK

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority of Taiwanese Application No. 095221311, filed on Dec. 4, 2006.

BACKGROUND OF THE INVENTION

1. Field of the Invention

The invention relates to a safety hook, more particularly to a safety hook with a safety lock unit.

2. Description of the Related Art

FIG. **1** illustrates a conventional safety hook **9** that includes an inverted J-shaped hook body **91** that defines a hook mouth, a latch unit **92** pivoted to the hook body **91** about a first axis (L**1**) below the hook mouth for blocking and unblocking the hook mouth, a safety lock unit **93** pivoted to the hook body **91** about a second axis (L**2**) above the first axis (L**1**) for preventing undesired unblocking movement of the latch unit **92**, and a releasing unit **94** that is pivoted to the hook body **91** about a third axis (L**3**) slightly above the second axis (L**2**) and that is pivotably connected to the safety lock unit **93**. The releasing unit **94** has a spring-biased operating end portion **941** for releasing the latch unit **92** from a locked state.

When the latch unit **92** is to be unlocked, the operating end portion **941** of the releasing unit **94** is pushed to rotate the releasing unit **94** about the third pivot axis (L**3**), which results in rotation of the safety lock unit **93** about the second pivot axis (L**2**) in the same direction to a releasing position that permits unlocking of the latch unit **92**. The latch unit **92** is formed with an operating protrusion **922** that is disposed opposite to the operating end portion **941** in order to permit the user to simultaneously press the operating end portion **941** of the releasing unit **94** and to move the latch unit **92** using one hand.

The conventional safety hook **9** is disadvantageous in that the releasing unit **94** has a relatively complicated structure. In addition, since the force applied to the operating protrusion **922** of the latch unit **92** is substantially perpendicular to the force applied to the operating end portion **941** of the releasing unit **94**, unlocking operation of the latch unit **92** is relatively inconvenient.

SUMMARY OF THE INVENTION

Therefore, the object of the present invention is to provide a safety hook that can overcome the aforesaid drawbacks associated with the prior art.

Accordingly, a safety hook of this invention comprises: a hook body defining a hook inner space therein and a hook mouth in spatial communication with the hook inner space; a latch unit including a latch plate that is pivoted to the hook body so as to be rotatable relative to the hook body between locked and unlocked positions, the latch plate having a blocking plate portion that blocks the hook mouth when the latch unit is disposed at the locked position, thereby denying access into the hook inner space; a releasing unit including a releasing plate that is pivoted to the hook body so as to be rotatable relative to the hook body, the releasing plate having a driving plate portion opposite to the blocking plate portion of the latch plate; and a safety lock unit including a bent safety plate that is disposed between the latch plate and the releasing plate and that is pivoted to the hook body so as to be rotatable relative to the hook body between restraining and releasing positions. The bent safety plate has a pivoting middle portion pivoted to the hook body, a restraining end portion extending from one end of the pivoting middle portion to abut against the blocking plate portion of the latch plate, and a driven end portion extending from the other end of the pivoting middle portion to abut against the driving plate portion of the releasing plate so as to permit the bent safety plate to be driven by the releasing plate to rotate from the restraining position to the releasing position. The restraining end portion of the bent safety plate is perpendicular to the blocking plate portion of the latch plate when the bent safety plate is disposed at the restraining position, thereby preventing rotation of the latch plate from the locked position to the unlocked position. The restraining end portion of the bent safety plate is inclined to the blocking plate portion of the latch plate at an angle less than 90 degrees when the bent safety plate is disposed at the releasing position, thereby permitting rotation of the latch plate from the locked position to the unlocked position.

BRIEF DESCRIPTION OF THE DRAWINGS

Other features and advantages of the present invention will become apparent in the following detailed description of the preferred embodiments with reference to the accompanying drawings, of which:

FIG. **1** is a schematic view of a conventional safety hook in a locked state;

FIG. **2** is an exploded schematic view of the first preferred embodiment of a safety hook according to the present invention;

FIG. **3** is an assembled schematic view of the first preferred embodiment in a locked and restrained state;

FIGS. **4** and **5** are schematic views to illustrate how a releasing unit is operated to release a latch unit from a safety lock unit of the first preferred embodiment;

FIG. **6** is an exploded schematic view of the second preferred embodiment of a safety hook according to the present invention;

FIG. **7** is an assembled schematic view of the second preferred embodiment in a locked and restrained state; and

FIGS. **8** to **10** are schematic views to illustrate how a releasing unit is operated to release a latch unit from a safety lock unit of the second preferred embodiment.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

Before the present invention is described in greater detail with reference to the accompanying preferred embodiments, it should be noted herein that like elements are denoted by the same reference numerals throughout the disclosure.

Referring to FIGS. **2** and **3**, the first preferred embodiment of a safety hook according to the present invention is shown to include: a hook body **1** defining a hook inner space **10** therein and a hook mouth **100** in spatial communication with the hook inner space **10**; a latch unit **2** including a latch plate **20** that is pivoted to the hook body **1** so as to be rotatable relative to the hook body **1** between locked and unlocked positions (see FIGS. **3** and **5**), the latch plate **20** having a blocking plate portion **21** that blocks the hook mouth **100** when the latch unit **2** is disposed at the locked position, thereby denying access into the hook inner space **10**; a releasing unit **4** including a releasing plate **40** that is pivoted to the hook body **1** so as to be rotatable relative to the hook body **1**, the releasing plate **40** having a driving plate portion **41** opposite to the blocking plate portion **21** of the latch plate **20**; and a safety lock unit **6** including a bent safety plate **60** that is disposed between the latch plate **20** and the releasing plate **40** and that is pivoted to

the hook body **1** so as to be rotatable relative to the hook body **1** between restraining and releasing positions (see FIGS. **3** and **4**). The bent safety plate **60** has a pivoting middle portion **62** pivoted to the hook body **1**, a restraining end portion **63** extending from one end of the pivoting middle portion **62** to abut against the blocking plate portion **21** of the latch plate **20**, and a driven end portion **64** extending from the other end of the pivoting middle portion **62** to abut against the driving plate portion **41** of the releasing plate **40** so as to permit the bent safety plate **60** to be driven by the releasing plate **40** to rotate from the restraining position to the releasing position. The restraining end portion **63** of the bent safety plate **60** is perpendicular to the blocking plate portion **21** of the latch plate **20** when the bent safety plate **60** is disposed at the restraining position, thereby preventing rotation of the latch plate **20** from the locked position to the unlocked position. The restraining end portion **63** of the bent safety plate **60** is inclined to the blocking plate portion **21** of the latch plate **20** at an angle less than 90 degrees when the bent safety plate **60** is disposed at the releasing position, thereby permitting rotation of the latch plate **20** from the locked position to the unlocked position.

In this embodiment, the restraining end portion **63** of the bent safety plate **60** and the driven end portion **64** of the bent safety plate **60** form an angle less than 180 degrees and greater than 90 degrees.

Referring again to FIGS. **2** and **3**, the hook body **1** has a base segment **11** with a front end **112**, and a hook end segment **12** with a free end **122** that cooperates with the front end **112** of the base segment **11** to define the hook mouth **100**. The latch plate **20** is pivoted to the front end **112** of said base segment **11**. The latch unit **2** further includes a first urging member **3** that urges the blocking plate portion **21** of the latch plate **20** to abut against the free end **122** of the hook end segment **12** of the hook body **1**.

The safety lock unit **6** further includes a second urging member **5** that urges the restraining end portion **63** of the bent safety plate **60** to abut against the blocking plate portion **21** of the latch plate **20**.

In this preferred embodiment, the latch plate **20** is pivoted to the hook body **1** about a first axis (A), the releasing plate **40** is pivoted to the hook body **1** about a second axis (B), and the bent safety plate **60** is pivoted to the hook body **1** about a third axis (C) that is parallel to the first and second axes (A, B). The second axis (B) is disposed between the first and third axes (A, C).

The base segment **11** is formed with a through-hole **110** adapted for extension of a cord or the like (not shown) therethrough, and further has a rear end **111**. The free end **122** of the hook segment is formed with a limiting protrusion **125**. The blocking plate portion **21** of the latch plate **20** of the latch unit **2** is formed with a top notch **210** for extension of the limiting protrusion **125** therethrough when the latch unit **2** is disposed at the locked position. The latch plate **20** is generally U-shaped and further has a pair of wing portions **22** extending transversely and respectively from opposite lateral edges of the blocking plate portion **21** and straddled on the hook body **1**. The latch unit **2** is pivoted to the hook body **1** through the wing portions **22**.

The releasing plate **40** of the releasing unit **4** is generally U-shaped and further has a pair of wing portions **42** extending transversely and respectively from opposite lateral edges of the driving plate portion **41** and straddled on the hook body **1**. The releasing unit **4** is pivoted to the hook body **1** through the wing portions **42**. The second axis (B,) is disposed adjacent to the rear end **111** of the base segment **11**. A protective flange **123** is formed on and protrudes outwardly from the hook end segment **12** so as to define a recess **121** thereat. The releasing plate **40** has a top end **43** that is bent outwardly and that has an outer edge which is disposed within the recess **121** defined by the protective flange **123** or which is flush with an outer edge of the flange **123** formed on the hook end segment **12** so as to prevent accidental movement of the releasing unit **4** from occurring when the latch unit **2** is to be retained at the locked position.

The bent safety plate **60** includes a connecting part **61** and a pair of wing parts **62'** extending transversely and respectively from two ends of the connecting part **61** and straddled on the hook body **1**. The wing parts **62'** cooperatively define the pivoting middle portion **62**, the restraining end portion **63**, and the driven end portion **64** of the bent safety plate **60**. A folded portion **65** is bent from one end of the driven end portion **64** of the bent safety plate **60**. The safety lock unit **6** is pivoted to the hook body **1** through the wing parts **62'**.

During unlocking operation of the safety hook, as shown in FIGS. **4** and **5**, the top end **43** of the releasing plate **40** is pressed such that there leasing unit **4** is pivoted about the second axis (B) in a first direction to drive the bent safety plate **60** to rotate about the third axis (C) in the first direction. As such, the restraining end portion **63** of the bent safety plate **60** is no longer perpendicular to the blocking plate portion **21** of the latch plate **20**, thereby permitting rotation of the latch plate **20**. At this time, a force exerted in a second direction opposite to the first direction is subsequently applied to a top end of the latch plate **21** to rotate the latch plate **20** about the first axis (A) so as to open the hook mouth **100**. The connecting part **61** abuts against the blocking plate portion **21** of the latch plate **20** when the hook mouth **100** is fully opened (see FIG. **5**).

As shown in FIGS. **6** and **7**, the second preferred embodiment of the safety hook according to this invention differs from the previous embodiment in that the third axis (C) is disposed between the first and second axes (A, B) and that the top end **43** of the releasing plate **40** is pivoted to the hook body **1**. Hence, the pressing action is to be applied to a bottom end of the releasing plate **40** (see FIG. **8**), and the rotating direction of the releasing plate **40** and the bent safety plate **60** of this preferred embodiment (see FIGS. **9** and **10**) is opposite to that of the previous embodiment (see FIGS. **4** and **5**) during unlocking operation of the latch unit **2**.

With the inclusion of the bent safety plate **60** in the safety lock unit **6** of the safety hook of this invention, the aforesaid drawbacks associated with the conventional safety hook can be eliminated.

While the present invention has been described in connection with what are considered the most practical and preferred embodiments, it is understood that this invention is not limited to the disclosed embodiments but is intended to cover various arrangements included within the spirit and scope of the broadest interpretation so as to encompass all such modifications and equivalent arrangements.

What is claimed is:

1. A safety hook comprising:

a hook body defining a hook inner space therein and a hook mouth in spatial communication with said hook inner space;

a latch unit including a latch plate that is pivoted to said hook body so as to be rotatable relative to said hook body between locked and unlocked positions, said latch plate having a blocking plate portion that blocks said hook mouth when said latch unit is disposed at the locked position, thereby denying access into said hook inner space;

a releasing unit including a releasing plate that is pivoted to said hook body so as to be rotatable relative to said hook body, said releasing plate having a driving plate portion opposite to said blocking plate portion of said latch plate; and

a safety lock unit including a bent safety plate that is disposed between said latch plate and said releasing plate and that is pivoted to said hook body so as to be rotatable relative to said hook body between restraining and releasing positions, said bent safety plate having a pivoting middle portion pivoted to said hook body, a restraining end portion extending from one end of said pivoting middle portion to abut against said blocking plate portion of said latch plate, and a driven end portion extending from the other end of said pivoting middle portion to abut against said driving plate portion of said releasing plate so as to permit said bent safety plate to be driven by said releasing plate to rotate from the restraining position to the releasing position, said restraining end portion of said bent safety plate being perpendicular to said blocking plate portion of said latch plate when said bent safety plate is disposed at the restraining position, thereby preventing rotation of said latch plate from the locked position to the unlocked position, said restraining end portion of said bent safety plate being inclined to said blocking plate portion of said latch plate at an angle less than 90 degrees when said bent safety plate is disposed at the releasing position, thereby permitting rotation of said latch plate from the locked position to the unlocked position;

wherein said restraining end portion of said bent safety plate and said driven end portion of said bent safety plate form an angle less than 180 degrees and greater than 90 degrees.

2. The safety hook as claimed in claim **1**, wherein said hook body has a base segment with a front end, and a hook end segment that cooperates with said front end of said base segment to define said hook mouth, said latch plate being pivoted to said front end of said base segment, said latch unit further including a first urging member that urges said blocking plate portion of said latch plate to abut against said hook end segment of said hook body.

3. The safety hook as claimed in claim **2**, wherein said safety lock unit further includes a second urging member that urges said restraining end portion of said bent safety plate to abut against said blocking plate portion of said latch plate.

4. The safety hook as claimed in claim **1**, wherein said latch plate is pivoted to said hook body about a first axis, said releasing plate being pivoted to said hook body about a second axis, said bent safety plate being pivoted to said hook body about a third axis that is parallel to said first and second axes, said second axis being disposed between said first and third axes.

5. A safety hook comprising:

a hook body defining a hook inner space therein and a hook mouth in spatial communication with said hook inner space;

a latch unit including a latch plate that is pivoted to said hook body so as to be rotatable relative to said hook body between locked and unlocked positions, said latch plate having a blocking plate portion that blocks said hook mouth when said latch unit is disposed at the locked position, thereby denying access into said hook inner space;

a releasing unit including a releasing plate that is pivoted to said hook body so as to be rotatable relative to said hook body, said releasing plate having a driving plate portion opposite to said blocking plate portion of said latch plate; and

a safety lock unit including a bent safety plate that is disposed between said latch plate and said releasing plate and that is pivoted to said hook body so as to be rotatable relative to said hook body between restraining and releasing positions, said bent safety plate having a pivoting middle portion pivoted to said hook body, a restraining end portion extending from one end of said pivoting middle portion to abut against said blocking plate portion of said latch plate, and a driven end portion extending from the other end of said pivoting middle portion to abut against said driving plate portion of said releasing plate so as to permit said bent safety plate to be driven by said releasing plate to rotate from the restraining position to the releasing position, said restraining end portion of said bent safety plate being perpendicular to said blocking plate portion of said latch plate when said bent safety plate is disposed at the restraining position, thereby preventing rotation of said latch plate from the locked position to the unlocked position, said restraining end portion of said bent safety plate being inclined to said blocking plate portion of said latch plate at an angle less than 90 degrees when said bent safety plate is disposed at the releasing position. thereby permitting rotation of said latch plate from the locked position to the unlocked position;

wherein said latch plate is pivoted to said hook body about a first axis, said releasing plate being pivoted to said hook body about a second axis, said bent safety plate being pivoted to said hook body about a third axis that is parallel to and that is disposed between said first and second axes.

* * * * *